

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

Slide Set 1: Introduction



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
- DO NOT REDISTRIBUTE
 - ▶ For use only by enrolled students in the course

Face Detection



Face Detection



Face Detection

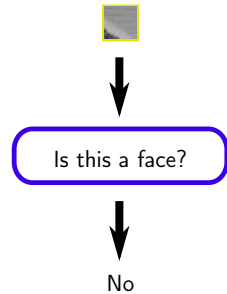


Is this a face?



No

Face Detection



Face Detection

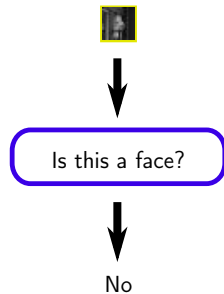


Is this a face?

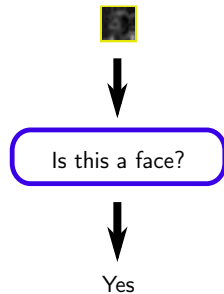


Yes

Face Detection



Face Detection



Face or Not Face?

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Face Detection

Face or Not Face?

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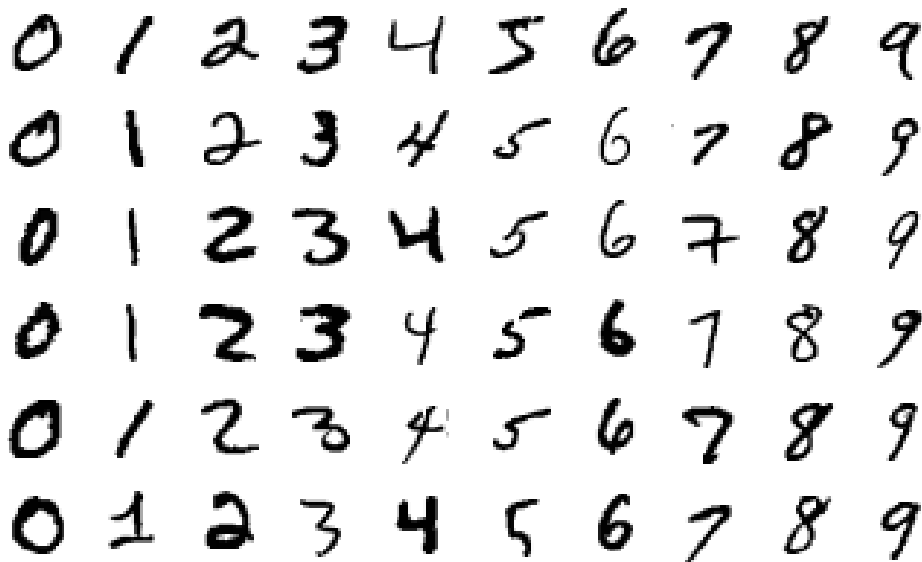
Face or Not Face?

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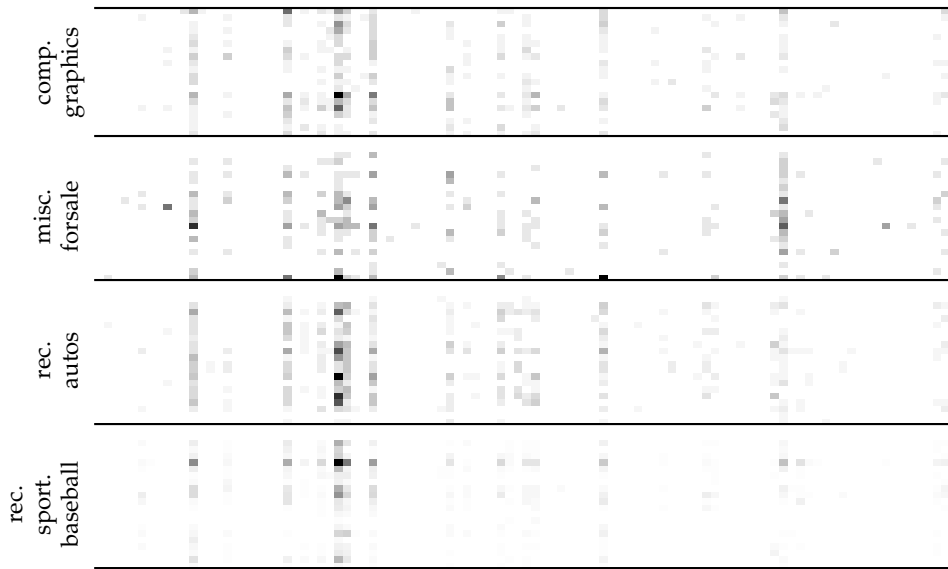
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Digit Recognition

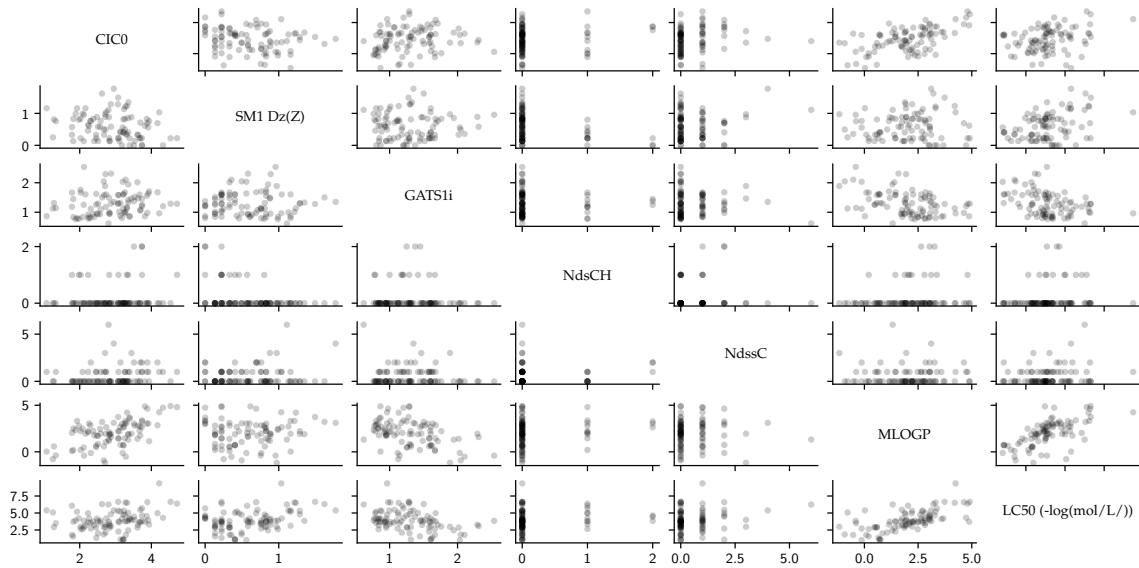


Document Topic Data



[Ken Lang, via Jason Rennie]

Fish Toxicity



Supervised Learning

The Data (**training set**):

- An (unordered) set of examples
- Each example has a set of **features** or **attributes**
 - ▶ binary (ex: is the word “health” present in the document or not?)
 - ▶ categorical (ex: relationship status)
 - ▶ real-valued (ex: weight)
- A **target** or **label** or **output**
 - ▶ binary (ex: has cancer)
 - ▶ categorical (ex: preferred brand of paper towels)
 - ▶ real-valued (ex: traffic delay in minutes)

Supervised Learning

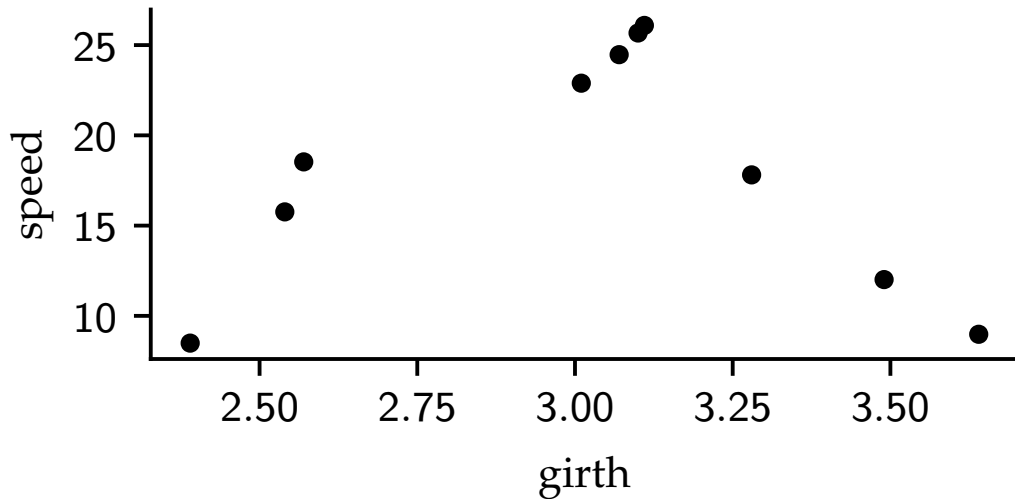
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The Goal:

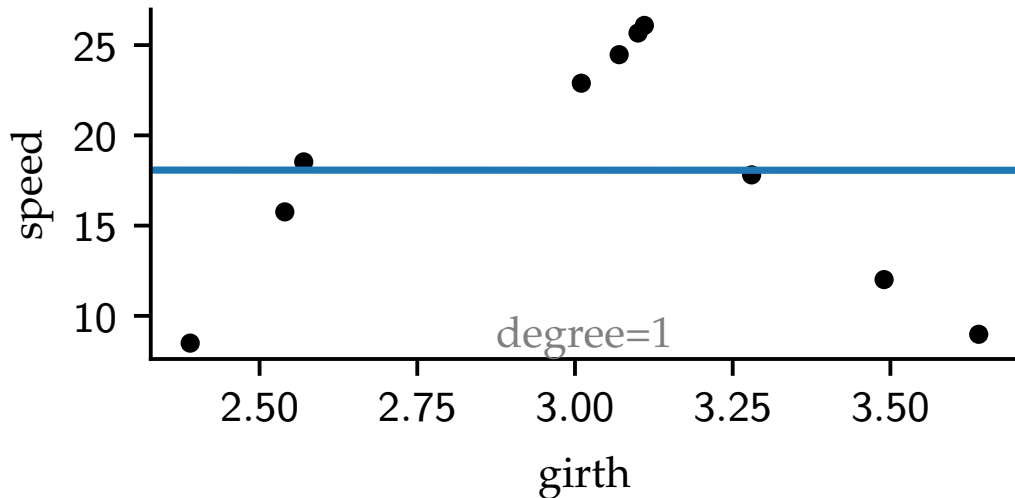
- Given the data above as the input to the learning algorithm,
- Produce a rule (**classifier** or **regressor**)
- that will map the feature values to the target
- The rule should work well on future, as-yet-unseen examples

Running Example



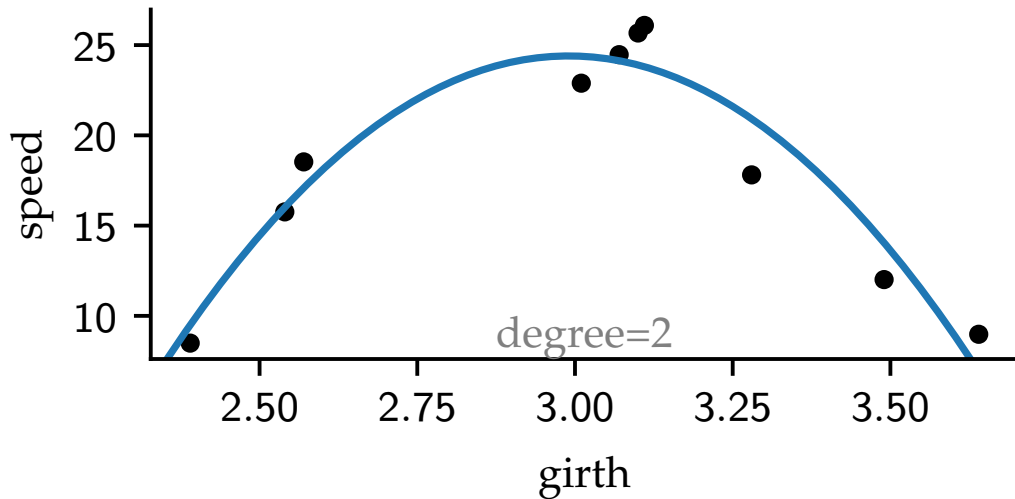
Hippopotamus land-speed data

Running Example



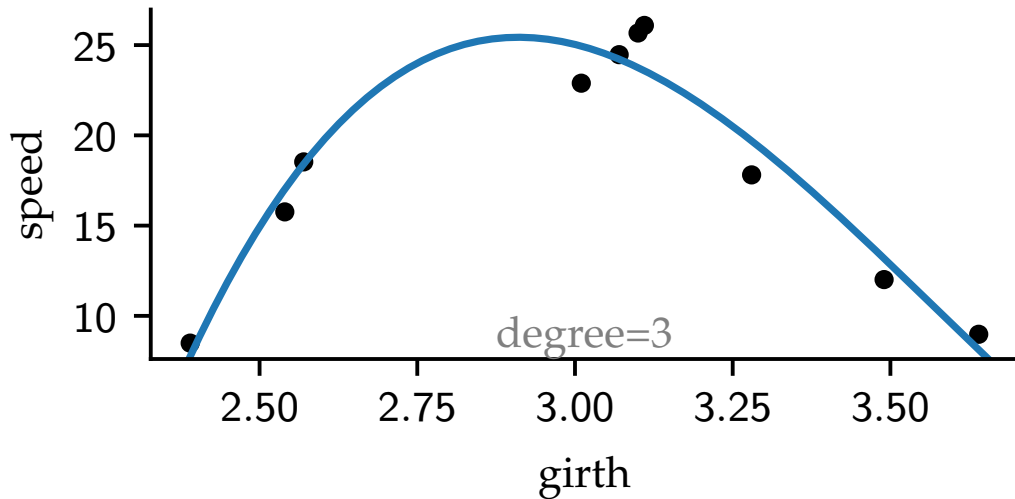
Hippopotamus land-speed data

Running Example



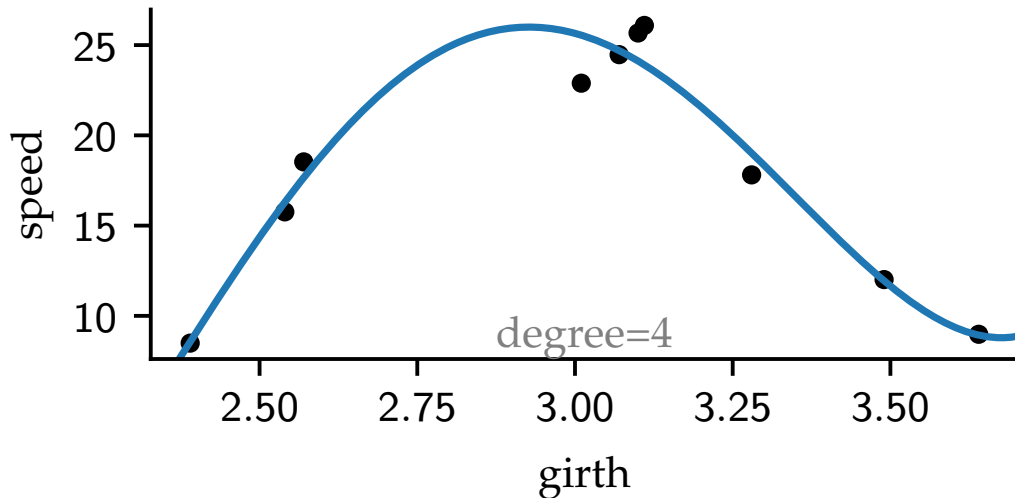
Hippopotamus land-speed data

Running Example



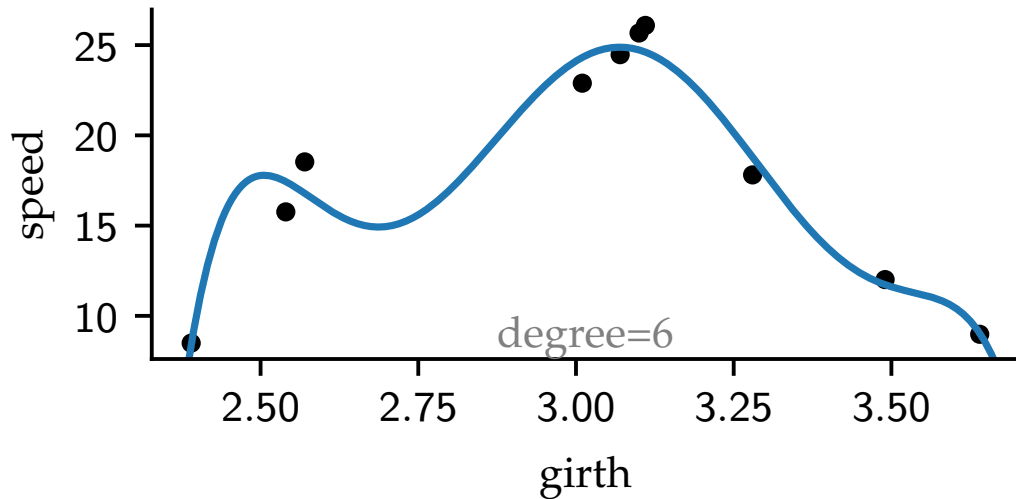
Hippopotamus land-speed data

Running Example



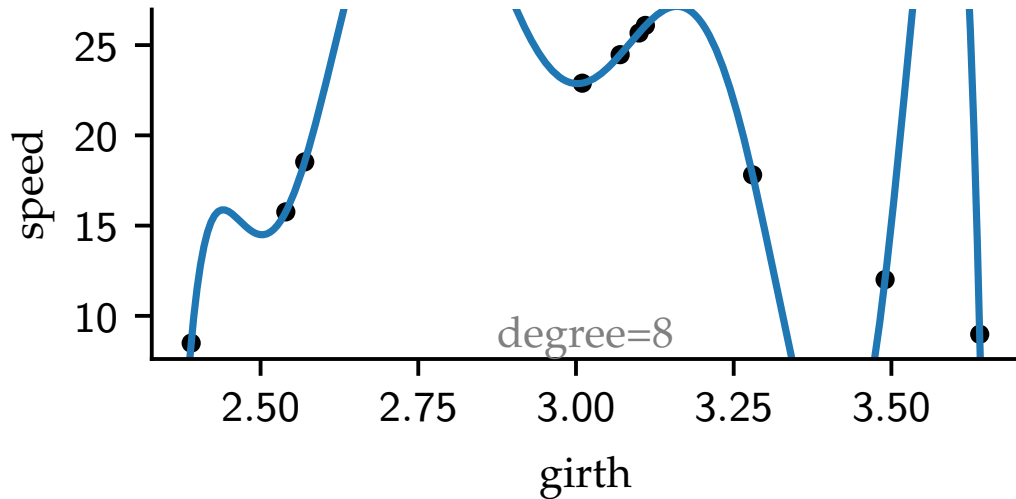
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Running Example



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Running Example



Hippopotamus land-speed data

Common Approach

Define the **hypothesis space**, the set of all possible output rules.

Examples:

- Degree-3 polynomials
- Linear dividing surfaces
- If-Then-Else chains

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- number incorrect (classification)
- sum of squared errors (regression)

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Examples:

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Develop an algorithm to

- pick the member of the hypothesis space that
- minimizes the error on the training set
- perhaps slightly modified

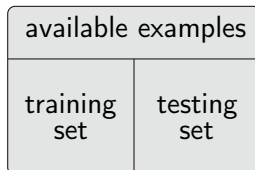
Testing Set

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Thus, we often use a **testing set**



Feed the training set into the learning algorithm.
Find the error of the resulting rule on the testing set.

Why is this hard?

- High dimensional spaces are weird
- Irrelevant attributes
- Redundant attributes
- Missing attributes
- Attribute noise
- Label noise

Terms defined

features or **attributes**: measured values associated with each example

target or **label** or **output**: desired output of the rule for each example

classifier, **regressor**: the rule that maps the features to the label

hypothesis space: the space of all rules that the machine learning algorithm is considering

training set: the data used as input to the machine learning algorithm

testing set: the data used to check the performance of the resulting rule